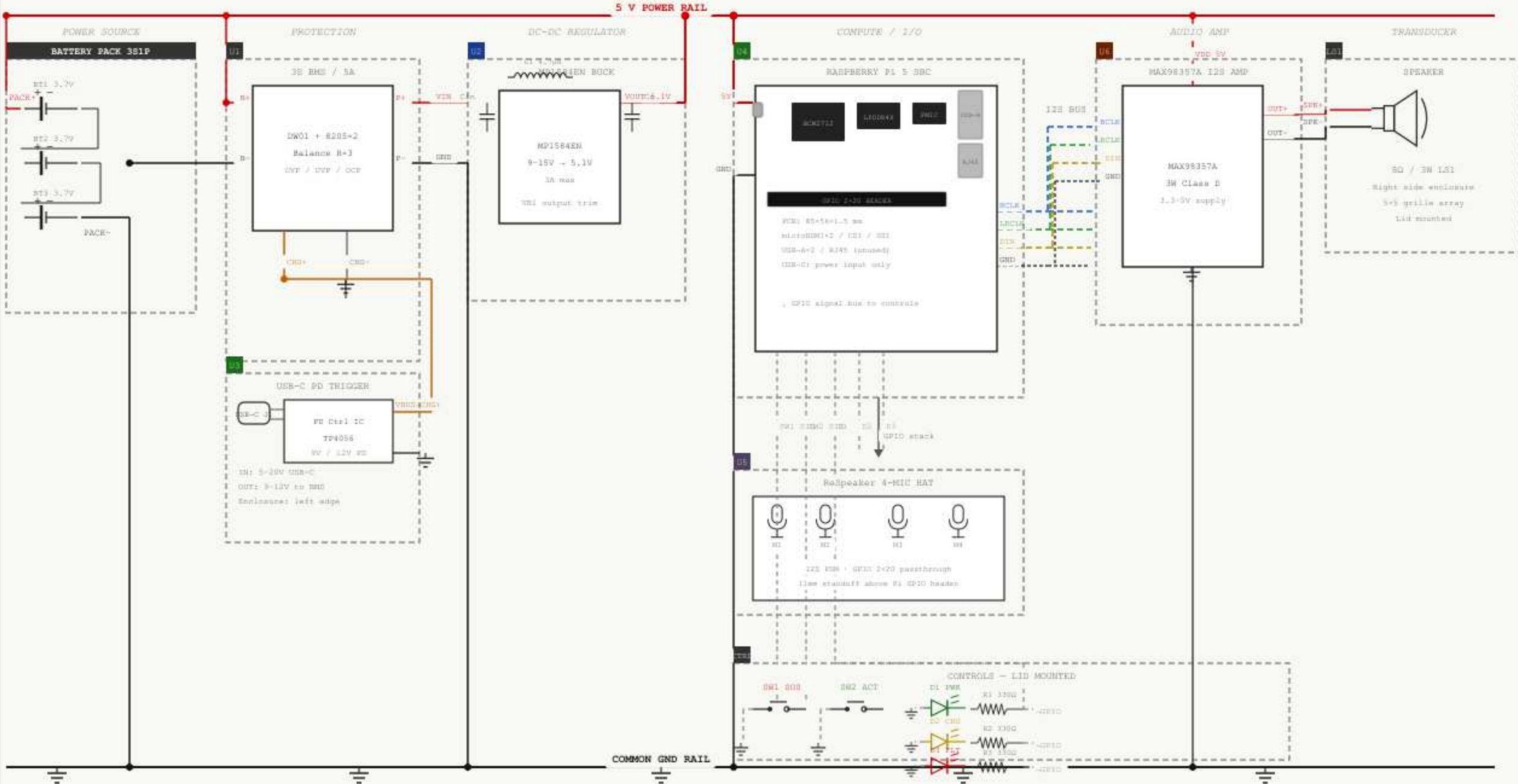




COMPONENT REFERENCE
BT1-BT3 18650 Li-Ion 3.7V 3Ah each - 381P pack = 11.1V nom / 12.6V full
U1 3S BMS 5A - DW01 protection IC + dual 8205 MOSFET - Balance R=3
U2 MP1584EN step-down converter - VIN 9-15V - VOUT 5.1V / 3A max
U3 USB-C PD trigger board - negotiates 9V or 12V PD contract
U4 Raspberry Pi 5 SBC - BCM2712 SoC / LPDDR4X / PMIC / GPIO 2x20
U5 RedSpeaker 4-Mic HAT - GPIO stack / 11mm standoff / I2S PCM bus
U6 MAX98357A I2S Class-D amplifier - 3W / 3.3V-5V supply
L1 8Q 3W speaker - enclosure right side, grille 5x5 array
SW1 Momentary push button - SWS (red cap) - lid mounted
SW2 Momentary push button - Action (green cap) - lid mounted
R1 PWR grn / R2 CHG yel / R3 FLT red - 3mm LEDs lid-mounted - R1-R3 330H

WIRE LEGEND
— PWR + (22 AWG red)
— GND (22 AWG black)
--- I2S BCLK (26 AWG blu)
--- I2S LCLK (26 AWG grn)
--- I2S DIN (26 AWG yel)
--- CHG VBUS (22 AWG org)
--- SIG / GPIO (26 AWG wht)
— SPS leads (24 AWG)
● Junction (connected)